Fatigue Risk of Bare Offshore Monopiles During Installation: A Python-Based Study

Technical summary | Riya Thripakkal Rajesh | M.Sc. ACCESS, TU Dresden | self-study portfolio project (not for design use)

Question. A monopile can stand bare for months before the turbine is connected. How much vibration and fatigue damage can it accumulate in a 6-month waiting period, and what should a battery-powered logger record to capture it?

Answer in short. (1) Waves alone do almost no fatigue damage to the bare monopile: 6-month Miner damage at the mudline weld is **3.3e-4** (Monte Carlo P5 to P95: 4.1e-5 to 2.7e-3), against an allowable of about 0.33 (design fatigue factor 3). (2) The damage is dominated by rare storms: sea states above Hs = 4.5 m are 4 % of the time but cause 47 % of the damage. (3) The suspicious stage is the tower without rotor: its critical wind speed (11.5 m/s) is in the normal wind range, so it sits in the vortex lock-in band for 1328 h of the 6 months. Whether that matters depends on the lock-in amplitude, which this study cannot predict: the mudline weld reaches D = 1/DFF only if the tower top oscillates with about 0.42 m amplitude (A/D = 0.09); a code-style screening estimate gives 0.09 m. (4) The logger therefore should be built to resolve very small accelerations (about 0.4 mg RMS in a quiet sea) and to catch storms and lock-in events, not to log everything.

1. Model and assumptions

Structure. Euler-Bernoulli beam finite-element model (1 m elements) of a reference monopile: D = 8.0 m, t = 90 mm, 30 m embedded, 30 m water depth, transition piece at 40 m, tapered tower (D 6.5 to 4.5 m) to a 140 m hub, 200 t platform and 450 t nacelle-and-rotor mass; added water mass on the wet length. Soil is a Winkler foundation with linearly increasing stiffness n_h (5 / 15 / 40 MN/m^3 for soft / medium / stiff). Four stages are analysed; first frequencies (medium soil) 1.22 / 0.86 / 0.51 / 0.26 Hz (soft to stiff: see table 1).

Loading. JONSWAP sea states (peak factor from the DNV rule), linear Airy waves, linearised Morison force (Cm = 2.0, Cd = 1.0, Borgman drag linearisation) with the MacCamy-Fuchs diffraction correction of the inertia term, because D/L reaches 0.14 to 0.34. The 8 m pile makes inertia dominate (drag below 0.5 % of the moment variance). The 6-month sea-state table is synthetic: Weibull Hs (scale 1.9 m, shape 1.35, mean Hs 1.7 m), lognormal Tp given Hs, steepness cut-off 1/15, 40 states.

Response and stress. Mudline overturning moment from the wave force (exact, quasi-static) plus a mode-1 dynamic correction (mode-acceleration form), damping 1 % of critical. Nominal bending stress at the mudline girth weld (Z = 4.37 m^3, SCF = 1). Time series are simulated from the spectrum (3 h, random phases) and counted by rainfall; damage uses the DNV-RP-C203 D-curve with thickness correction (t/25 mm)$^{0.2}$ and Miner's rule. Spectral damage (Dirlik and narrow-band) is the cross-check.

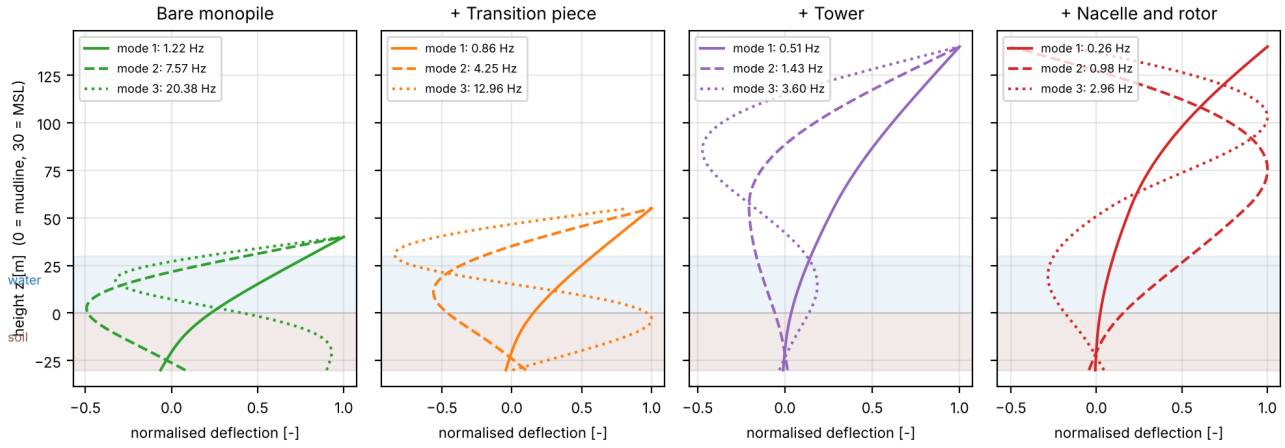


Figure 1. Mode shapes 1 to 3 for the four installation stages (medium soil). Only the three bare stages are free of the heavy rotor mass; the rotor stage has its first mode at 0.26 Hz, just above the wave band.

2. Results

Table 1. Stage comparison (baseline S-N curve: D, seawater with cathodic protection). Allowable damage is 1/DFF = 0.33. The rotor stage is shown for completeness: its wave response is qualitative only (no aerodynamic damping, JONSWAP tail).

Stage	f_1 soft / med / stiff [Hz]	Mudline stress RMS, LC5 [MPa]	6-month damage, waves (Dirlik)	Monte Carlo waves P5 / P50 / P95	Time in VIV band [h]
Bare monopile	0.82 / 1.22 / 1.54	4.5	3.3e-4	4.1e-5 / 3.5e-4 / 2.7e-3	< 1 (Phase B)
+ Transition piece	0.61 / 0.86 / 1.04	4.5	3.6e-4	4.5e-5 / 3.9e-4 / 3.0e-3	< 1 (Phase B)
+ Tower	0.40 / 0.51 / 0.57	4.8	5.1e-4	6.5e-5 / 5.8e-4 / 4.7e-3	1328
+ Nacelle and rotor	0.22 / 0.26 / 0.27	9.5	2.3e-2	9.1e-4 / 1.5e-2 / 4.0e-1	728

Waves. The three bare stages respond almost quasi-statically (dynamic amplification 1.01 to 1.2, Phase B), so the mudline stress follows the wave moment: LC5 (Hs 4 m, Tp 8 s) gives 4.5 MPa RMS and about 18 MPa maximum in 3 h, consistent with

the hand estimate M_{rms}/Z = 4.4 MPa. At such low stress ranges nearly every cycle lies below the knee of the S-N curve, so the result is controlled by the shallow-slope branch and by the choice of curve: in air and in seawater with cathodic protection give identical damage (3.3e-4); free corrosion gives 5.3e-3 (about 16 times higher), which is still only 2 % of the allowable. Rainflow counting, Dirlik and the narrow-band formula agree within 4 % for all stages.

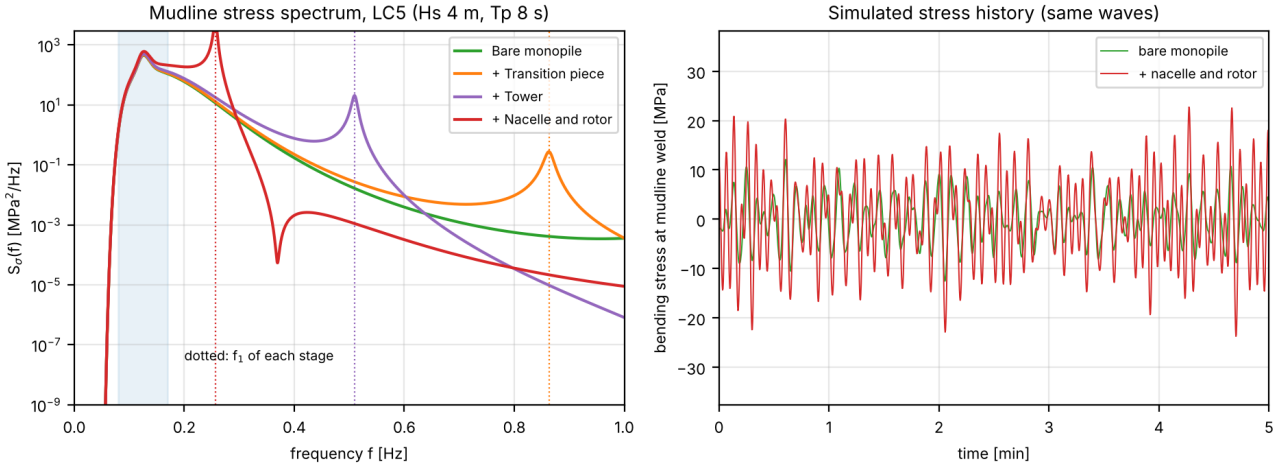


Figure 2. Response spectrum of the mudline bending stress for LC5 (left) and a simulated 5-minute stress history (right). The bare pile follows the wave band; the tower and the rotor stage show their own resonances.

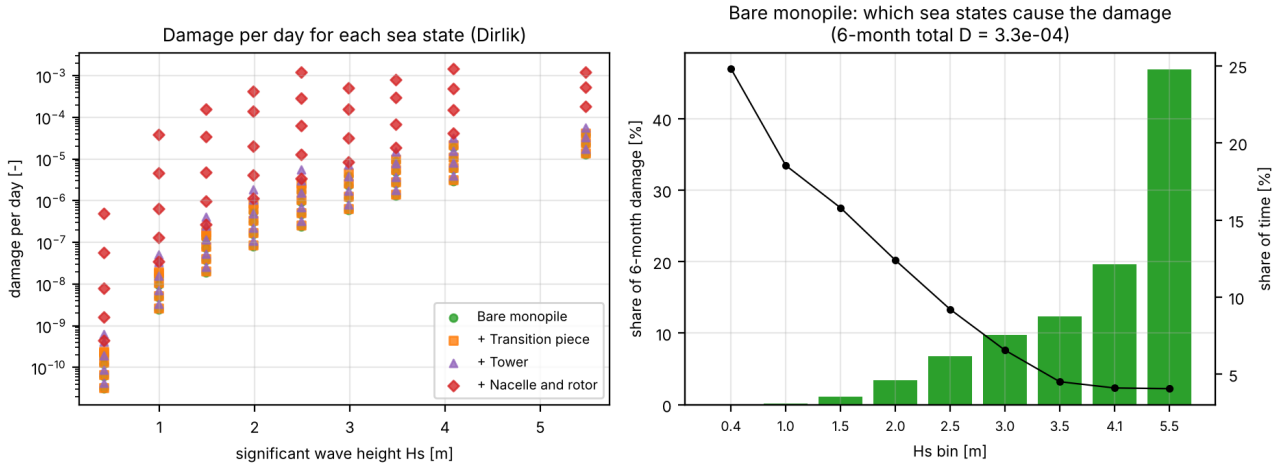


Figure 3. Damage per day for each sea state (left) and the share of the 6-month damage by Hs bin for the bare monopile (right, with the share of time). Storms drive the damage; calm seas contribute almost nothing.

3. Vortex-induced vibration: a scenario, not a prediction

Phase B showed that the bare pile and the pile with transition piece have critical wind speeds of 36 m/s and above, so VIV is not a concern there. The tower without rotor (f_1 = 0.51 Hz, U_c = 11.5 m/s) is inside a +/-20 % lock-in band for about 1328 h; with the rotor mounted but not operating, 728 h. Exposure is not damage. A linear model cannot predict the lock-in amplitude, so the scenario is built the other way round: the stress range at the mudline is 2 A m/Z (m = 322 MNm of mudline moment per metre of top amplitude), cycles are counted at f_1 over the time in the band with a triangular amplitude shape, and damage is plotted against the assumed amplitude (Figure 4). The reference point is an EN 1991-1-4 Annex E style estimate (Scruton number 19, A = 0.09 m); its constants must be verified against the standard. At the mudline the damage at that amplitude is 1.7e-4; the tower-base weld at 55 m is about 3 times lower in this mode (4.9e-5). The amplitude that would use up the allowable damage (D = 1/DFF) is 0.42 m for the tower stage and 0.80 m for the rotor stage.

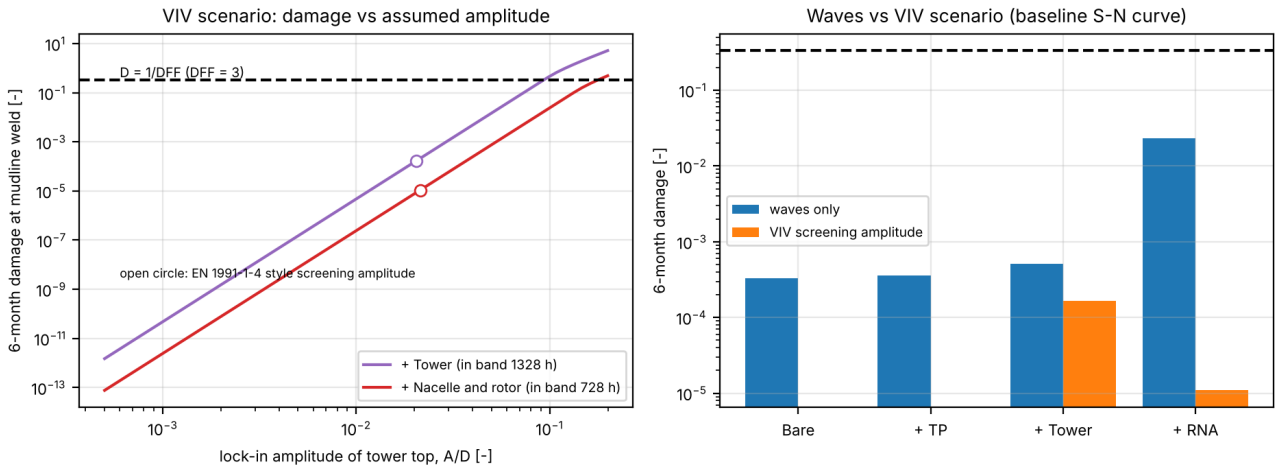


Figure 4. Left: 6-month damage at the mudline weld against the assumed lock-in amplitude. Right: waves versus the VIV screening scenario by stage. Log scales.

4. Uncertainty (Monte Carlo)

1500 samples of: soil stiffness (lognormal, median 15 MN/m^3, 5th to 95th percentile 5.6 to 40), damping (median 1.5 %, 0.4 to 4 %), a wave-severity factor on the whole scatter table (lognormal, sigma 0.25), and for VIV an amplitude model factor (lognormal, sigma 0.5) and the fraction of band time with developed lock-in (uniform 0.25 to 1). Input distributions are judgement, not data. Bare monopile: median 3.5e-4, P95 2.7e-3, P99 6.5e-3, so even the 99th percentile is 2 % of the allowable. Tower stage with waves and the VIV scenario: median 7.5e-4, P95 8.2e-3, P99 3.7e-2, probability of exceeding 1/DFF 0.3 %. The spread is driven by wave severity (rank correlation 0.82) before the VIV amplitude factor (0.27) and damping (-0.26); soil stiffness hardly matters for the bare stages. The parked rotor stage reaches P95 = 4.0e-1 because its resonance sits on the spectrum tail; treat that number as a warning that aerodynamic damping and the tail shape must be modelled before any conclusion.

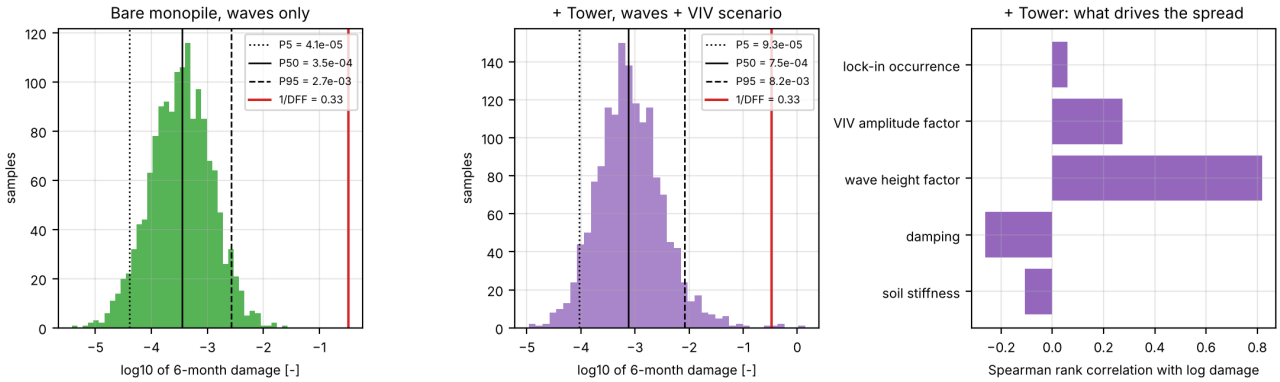


Figure 5. Distribution of the 6-month damage (log scale) for the bare monopile (waves) and the tower stage (waves plus VIV scenario), and rank correlations of the inputs with the tower-stage damage.

5. Logger recommendation

Table 2. Requirements derived from the response results (signal levels from mode 1 only, so a lower bound for high-frequency content).

Requirement	Value	Why (from the results)
Measured quantity	2 horizontal accelerations at the highest accessible point; inclination and temperature as slow channels; one strain or mudline channel if access allows	Mode-1 motion is largest at the top; frequencies, damping and lock-in show up in acceleration
Range	+/- 1 g (selectable)	Wave 3-h peak 55 mg (tower stage); lock-in at D = 1/DFF: 437 mg; parked rotor stage 180 mg
Noise floor	at most 29 ug/sqrt(Hz) in 0.03 to 2 Hz	Smallest sea state that still matters (Hs about 2.0 m, 99 % of damage) gives only 0.4 mg RMS on the bare pile; noise set to 10 % of it
Frequency band	0.03 to 7 Hz	Wave band 0.08 to 0.17 Hz; bare-pile mode 1 at 0.8 to 1.5 Hz; mode 2 at 7.6 Hz (8.8 Hz stiff soil)
Sampling rate	25 Hz raw bursts, 10 Hz for the always-on channel	2.5 times the bare-pile mode 2 (stiff soil); the stress-relevant content is below 2 Hz
Stored quantities	10-min mean, RMS, max, min, dominant frequency and a 32-band spectrum; hourly rainflow range histogram (32 bins); raw bursts on triggers	Statistics keep spectra and exposure, the histogram keeps damage, bursts keep waveforms

Table 3. Logging strategies. Device values are assumptions (sensor 3 mW, microcontroller 30 mW active and 0.03 mW asleep, 1 uJ per stored byte, 24-bit, 2 channels); check them against data sheets. Memory differs by three orders of magnitude, but energy hardly differs, because the always-on sensor dominates the budget.

Strategy (6 months)	Memory [MB]	Mean power [mW]	Energy [Wh]	Keeps / loses
Raw continuous (25 Hz, 2 ch)	2365	6.2	27	everything: spectra, damping, VIV, cycles / nothing, but memory and download
10-min statistics + 32-band spectrum	3.2	4.2	19	RMS, max, spectrum, dominant frequency / cycle sequence; transient events
On-board rainflow histogram (stress range)	3.9	5.1	22	damage directly; cycle counts per hour / raw waveform; needs calibrated moment-to-acceleration transfer function
Event-triggered raw bursts + statistics (storm trigger, bare pile: 13 % of time)	304	4.6	20	full waveform where it matters + background statistics / raw data outside trigger; trigger-threshold risk
Event-triggered raw bursts + statistics (lock-in trigger, tower stage: 30 % of time)	720	5.2	23	full waveform where it matters + background statistics / raw data outside trigger; trigger-threshold risk

Recommendation. Run a hybrid: always-on 10-minute statistics with a spectrum, an on-board hourly rainfall histogram once the moment-to-acceleration transfer function has been calibrated against the model, and event-triggered raw bursts at 25 Hz when the 10-minute RMS or the wind speed enters a lock-in band or a storm. This keeps the damage record, the spectrum and the waveforms where they matter at a few hundred MB instead of 2.4 GB, and one primary lithium cell is enough for 6 months in this model. The first thing to measure is the lock-in amplitude of the tower stage, because it is the one unknown that this study cannot close.

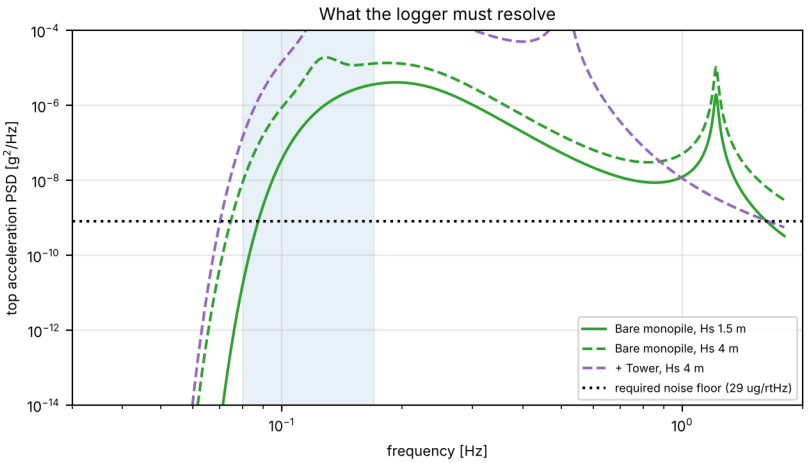


Figure 6. Top-acceleration spectra of the bare pile and the tower stage against the required sensor noise floor and the wave band.

6. Limitations and verification

Limitations. Linear soil without cyclic degradation; one mode only (higher modes matter for the rotor stage); 1 % constant damping and no aerodynamic damping; Airy waves with no stretching, breaking, slamming, current or marine growth; generic synthetic sea states and Weibull wind, not site data; Strouhal number 0.2 and a +/-20 % band although the Reynolds number is about 3×10^6, where shedding is less coherent and the tapered tower locks in only partly; VIV amplitude is an assumption; nominal stress with SCF = 1 at one cross-section (the maximum moment is below the seabed, and flanges and attachments are not checked); DNV-RP-C203 constants, the thickness exponent and the EN 1991-1-4 constants were entered from memory and **must be verified against the current editions before any number is reused**. Wind elevations in Phase B are measured from the mudline rather than from still-water level, which makes wind speeds at the tower top about 3 % high; this does not change the conclusions. The damping, soil and wave factors in the Monte Carlo are judgement.

Verification. checks.py reproduces the Phase B moments to 1e-13, the four Phase A frequencies, the critical wind speed, matches an independent rainfall implementation, and shows time-domain and spectral damage within 4 %. A latent normalisation issue in fem.modes() (rotation degrees of freedom left unscaled) changes the generalised masses by less than 1 tonne and does not affect Phase B; it is bypassed in modal.py.